Advanced Algorithms (Quiz 3)

Course Code: CSE426
Total Marks: 20
Section 2

ID: ________________ **Name:** ____________________ **Serial:** ________

Questions

Q1. **(10 Marks)** Prove the following statement,"If G is a connected graph with n vertices and no cycles, then G has (n-1) edges."

Q2. **(10 Marks)** Prove the following statement,"If G is a connected graph with n vertices and (n-1) edges then G has no cycle"

Q3. **(1+1 Marks)** A graph G has 2 disjoint cycles. The first cycle has 6 edges in it and the second one has 11 edges. How many edges do we need to remove to make the graph into a tree? How many ways can we do that?.

Q4. **(3 Marks)** Let's say you draw n full binary trees. The first tree has 0 internal vertices , the second one has 1 internal vertices and so on. The last tree has $(n-1)$ internal vertices. Now derive a formula that outputs the sum of all vertices in those n full binary trees.